

=> Laplace approximation

=> Goal:

1. $E[a(\theta)] = \frac{\int \dots \int a(\theta) f(\theta) d\theta}{\int f(\theta) d\theta}$

2. $a(\theta)$ is a function of θ

3. $f(\theta)$ is the unnormalized posterior

e.g. $f(\theta) = L(\theta) \pi(\theta)$, penalized likelihood

=> Let $h(\theta) = -\log f(\theta)$ [negative log unnorm. post.]

$$f(\theta) = e^{-h(\theta)}$$

when, sample size n is very large

$$\theta \xrightarrow{\text{approximation}} N(\hat{\theta}_0, \hat{\Sigma}_0)$$

=> Work details:

$$\text{Let } \theta_0 = \arg \min_{\theta} h(\theta)$$

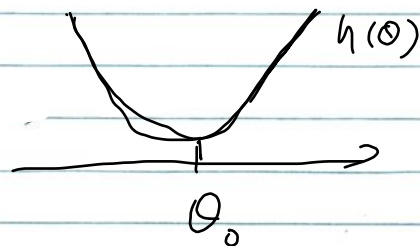
By Taylor expansion

$$h(\theta) = h(\theta_0) + \nabla h(\theta_0)^T (\theta - \theta_0)$$

$$+ \frac{1}{2} (\theta - \theta_0)^T \nabla^2 h(\theta_0) (\theta - \theta_0)$$

$$+ o((\theta - \theta_0)^2)$$

up to
using 2nd order



$$h(\theta) \approx \tilde{h}(\theta) = h(\theta_0) + \frac{1}{2}(\theta - \theta_0)^T \nabla^2 h(\theta_0) (\theta - \theta_0)$$

Based on the above $\tilde{h}(\theta)$

$$f(\theta) \approx \tilde{f}(\theta) = e^{-\tilde{h}(\theta)}$$

$$= e^{-h(\theta_0)} \cdot e^{-\frac{1}{2}(\theta - \theta_0)^T \nabla^2 h(\theta_0) (\theta - \theta_0)}$$

$$= \underbrace{f(\theta_0) \cdot (2\pi)^{\frac{p}{2}} |\nabla^2 h(\theta_0)|^{-\frac{1}{2}}}_{\Delta} \underbrace{(2\pi)^{-\frac{p}{2}} |\nabla^2 h(\theta_0)|^{\frac{1}{2}}}_{\Delta} e^{-\frac{1}{2}(\theta - \theta_0)^T \nabla^2 h(\theta_0) (\theta - \theta_0)}$$

$$\Delta \longrightarrow \int \Delta d\theta = 1$$

[Review: if $X \sim N_p(\mu, \Sigma)$, then

$$\text{PDF of } X: p(x) = (2\pi)^{-\frac{p}{2}} |\Sigma|^{-\frac{1}{2}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)}$$

That is:

$$1) \theta \sim N(\theta_0, \nabla^2 h(\theta_0)^{-1})$$

$$2) \int f(\theta) d\theta \approx f(\theta_0) (2\pi)^{\frac{p}{2}} |\nabla^2 h(\theta_0)|^{-\frac{1}{2}}$$

example: some special case.

$$1) f(\theta) = L(\theta, D) \pi(\theta)$$

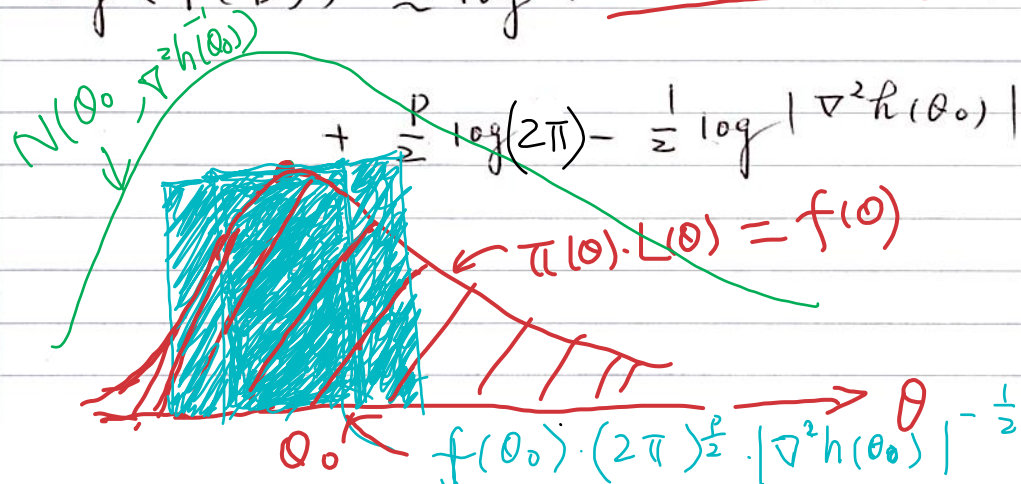
$$P(D) = \int L(\theta; D) \pi(\theta) d\theta$$

$$\log(P(D)) \approx \log(\underbrace{P(D|\theta_0)}_{\text{green curve}} \underbrace{\pi(\theta_0)}_{\text{red curve}})$$

$$+ \frac{p}{2} \log(2\pi) - \frac{1}{2} \log |\nabla^2 h(\theta_0)|$$

$$\pi(\theta) \cdot L(\theta) = f(\theta)$$

$$f(\theta_0) \cdot (2\pi)^{\frac{p}{2}} |\nabla^2 h(\theta_0)|^{-\frac{1}{2}}$$



=> A simplification of this formular

is could Bayesian Information

=> criterion $\sigma^2 h(\theta_0) \approx n I(\theta_0) \approx \begin{pmatrix} n & & \\ & \ddots & \\ & & n \end{pmatrix}$

Assuming $\sigma^2 h(\theta_0)$ is diagonal

$$|\sigma^2 h(\theta_0)| \approx n^p$$

$$\text{BIC} = -2 \left[\log(P(D|\theta_0) \pi(\theta_0)) + \frac{p}{2} \log(2\pi) - \frac{1}{2} p \log n \right]$$

↙

$$\approx -2 \log P(D|\theta_0) + [p \log n - \log(2\pi)^p]$$

$$\approx -2 \log P(D|\theta_0) + p \log n$$

$$\underline{-2 \log L(\theta_0) + p \log n}$$

AIC:

$$-2 \log L + 2p$$

2) example 2: second case:

$$a(\theta) = A^T \theta + a_0$$

$$E_p(a(\theta)) \approx E_{\tilde{p}}(a(\theta))$$

$$= A^T E_{\tilde{p}}(\theta) + a_0$$

$$= A^T \theta_0 + a_0$$

3) For general, $a(\theta)$, we can generate

$$\theta_1, \dots, \theta_N \sim N(\theta_0, \sigma^2 h^{-1}(\theta_0))$$

$$\text{then, } \hat{E}(a(\theta)) = \sum_{i=1}^N a(\theta_i) / N$$